

## Wazuh VirusTotal Integration for Malware Detection

By integrate Wazuh with VirusTotal we can take our system security to the next level.

Whenever a new file is created or modify, Wazuh will send the file hash to VirusTotal to check if it is malicious virus or malware for analysis.

### 1- Protecting Data Confidentiality via Hash-Only Scans

The very Important point is that: Wazuh never upload your actual file to VirusTotal, it only matches the hash, because if wazuh upload the file the VirusTotal it will breach basic security principle, specially CIA Triad (confidentiality, Integrity and available)

### 2- Why integration is so important

Integration is so important because it automatic threat hunting, so, we don't have to manually check every file, wazuh will do all this automatically. With real time alert, if a malicious file is detecting, will get an immediate alert

### 3- Enable our VirusTotal feature

Go to our wazuh server or manager configuration file (/var/ossec/etc/ossec.conf) using this command: `sudo nano /var/ossec/etc/ossec.conf` and at the end of the file, past this:

```
<ossec_config>
<integration>
  <name>virustotal</name>
  <api_key>Your_VirusTotal_api_key</api_key>
  <group>syscheck</group>
  <alert_format>json</alert_format>
</integration>
</ossec_config>
```

And save

**NB:** To get Your\_VirusTotal\_api\_key, create an account first here <https://www.virustotal.com>

And your api\_key will look like this



#### 4- Create a local rule in `local_rules.xml` file

Local rule is a configuration file that tells Wazuh to trigger the VirusTotal scan feature whenever an `.exe`, `.dll`, or `.ps1` file is created.

Go to Rules: by using this command: `sudo nano /var/ossec/etc/rules/local_rules.xml`

And add this configuration block at the end:

```
<group name="sysmon,virustotal,">
  <!-- This rule will fire when a new .exe or .dll file is created -->
  <rule id="100100" level="7">
    <if_sid>61605</if_sid> <!-- This is the Sysmon Event ID for FileCreate -->
    <field name="win.eventdata.targetFilename">\\.exe$|\\.dll$|\\.ps1$</field>
    <description>File - New executable or script created -> VirusTotal Scan Triggered</description>
    <group>virustotal,</group>
  </rule>
</group>
```

And save

After that restart our Wazuh server. Do `sudo systemctl restart wazuh-manager.service`

**This:**

```
<field name="win.eventdata.targetFilename">\.exe$|\.dll$|\.ps1$</field>
```

**Means that:** whenever an .exe, .dll, or .ps1 file is created, it must trigger the VirusTotal scan.

## 5- Simulate and attack

Go to your windows machine, and download a malware file here <https://www.eicar.org/download-anti-malware-testfile/> and save it

**NB:** Chose testfile and save that malware file in a directory you have already configured with File Integrity Monitoring

Now go to your wazuh SIEM, and you can see the log start to generate, and VirusTotal has detected the malware file



